

Functional Test for ESP32 Rainbow PCB Assembly

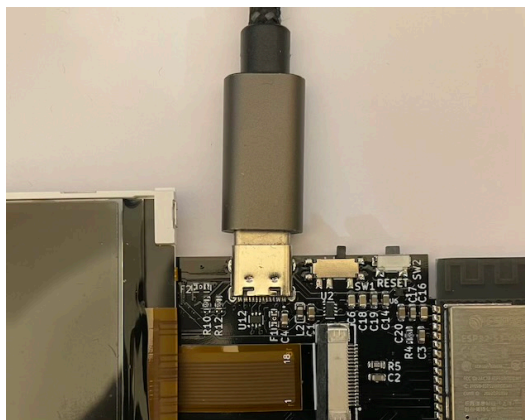
Required Equipment

- USB-C cable
- USB Power supply

Test Procedure

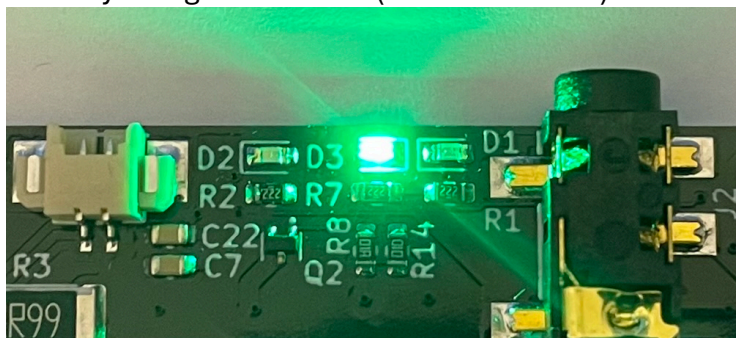
Step 1: Connect Over USB-C

1. Plug a USB-C cable into the PCB's USB-C connector.
2. Connect the other end of the cable to a computer.



Step 2: Confirm Green LED D1 is Lit

1. Verify that green LED D3 (Power Indicator) is illuminated.

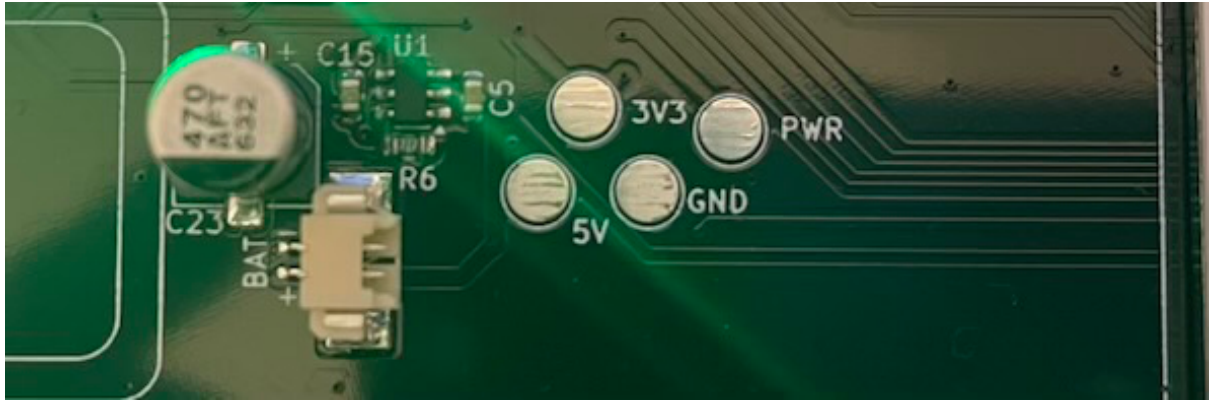


Pass Criteria:

- D3 is ON → Proceed to Step 3

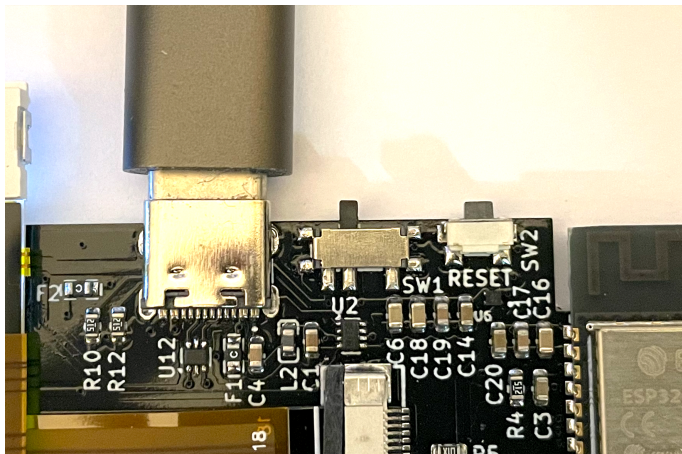
Troubleshooting

- D3 is OFF → Check soldering on USB connector J1, ICs: U12, U2. Check test points 5v and PWR. Measured voltage should be 5v



Step 3: Move Switch SW1 to Left Position

1. Locate switch SW1 on the PCB.
2. Move the switch to the left position.



Step 4: Confirm Green LED D2 is Lit

1. Verify that LED D2 is illuminated after moving the switch.

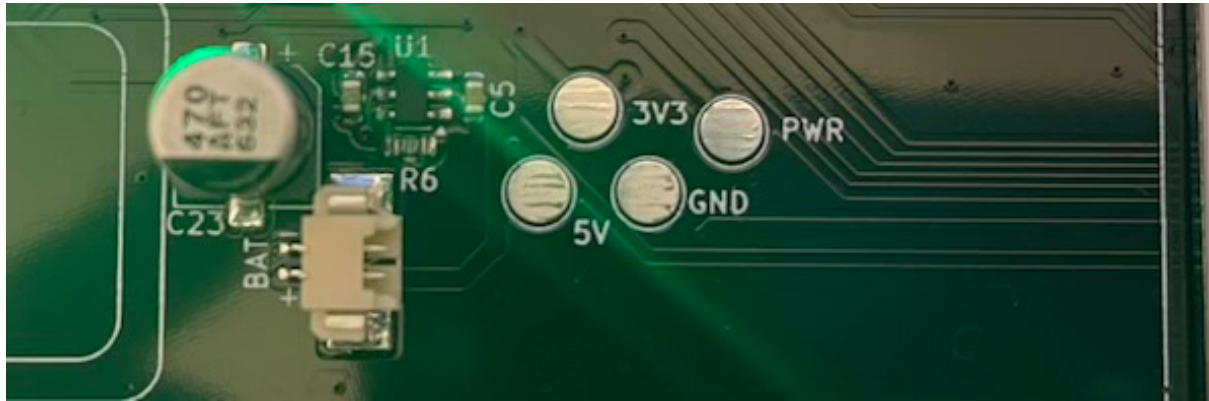


Pass Criteria:

- D2 is ON → Passed

Troubleshooting

- D2 is OFF → Check switch position. Check soldering on U6. Check test point 3V3



Step 4 Confirm USB device is present on PC

USB device should appear on the PC.

Pass Criteria:

- Device appears on the PC

Troubleshooting

- Check soldering on USB connector
- Check soldering on ESP32
- Check soldering on U12